

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396863
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Pimenta; Luiz et al.

Methods and apparatus for performing spine surgery

Abstract

Systems and methods are described for correcting sagittal imbalance in a spine including instruments for performing the controlled release of the anterior longitudinal ligament through a lateral access corridor and hyper-lordotic lateral implants.

Inventors: Pimenta; Luiz (Sao Paulo, BR), Morris; Andrew (San Diego, CA), Snider; Brian (San Diego, CA), Serra; Michael (San Diego, CA)

Applicant: NuVasive, Inc. (San Diego, CA)

Family ID: 1000008779734

Assignee: Globus Medical Inc. (Audubon, PA)

Appl. No.: 17/647226

Filed: January 06, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220125597 A1	Apr. 28, 2022

Related U.S. Application Data

continuation parent-doc US 16545821 20190820 US 11246713 child-doc US 17647226
continuation parent-doc US 15702685 20170912 US 10426627 20191001 child-doc US 16545821
continuation parent-doc US 14918137 20151020 US 9757246 20170912 child-doc US 15702685
continuation parent-doc US 14578215 20141219 US 9192482 20151124 child-doc US 14918137
continuation parent-doc US 13653335 20121016 US 8920500 20141230 child-doc US 14578215
continuation parent-doc US 12799021 20100416 US 8287597 20121016 child-doc US 13653335
us-provisional-application US 61319823 20100331
us-provisional-application US 61212921 20090416

Publication Classification

Int. Cl.: **A61F2/44** (20060101); **A61B17/02** (20060101); **A61B17/32** (20060101); **A61B17/3209** (20060101); **A61B17/3211** (20060101); **A61B17/34** (20060101); **A61F2/30** (20060101); **A61F2/46** (20060101); A61B17/00 (20060101); A61B90/00 (20160101); A61F2/28 (20060101)

U.S. Cl.:

CPC **A61F2/442** (20130101); **A61B17/0206** (20130101); **A61B17/320016** (20130101); **A61B17/3209** (20130101); **A61B17/3211** (20130101); **A61B17/3421** (20130101); **A61F2/30771** (20130101); **A61F2/4455** (20130101); **A61F2/447** (20130101); **A61F2/4611** (20130101); A61B2017/00261 (20130101); A61B2017/0256 (20130101); A61B2017/0262 (20130101); A61B17/320036 (20130101); A61B2017/320052 (20130101); A61B2090/08021 (20160201); A61B2090/0817 (20160201); A61F2002/2817 (20130101); A61F2002/2835 (20130101); A61F2002/30062 (20130101); A61F2002/30176 (20130101); A61F2002/30266 (20130101); A61F2002/30281 (20130101); A61F2002/30433 (20130101); A61F2002/30576 (20130101); A61F2002/30578 (20130101); A61F2002/30593 (20130101); A61F2002/30607 (20130101); A61F2002/30616 (20130101); A61F2002/30772 (20130101); A61F2002/30774 (20130101); A61F2002/30777 (20130101); A61F2002/30784 (20130101); A61F2002/30785 (20130101); A61F2002/30787 (20130101); A61F2002/30828 (20130101); A61F2002/30904 (20130101); A61F2002/3093 (20130101); A61F2002/444 (20130101); A61F2310/00011 (20130101); A61F2310/00179 (20130101); A61F2310/00359 (20130101)

Field of Classification Search

CPC: A61B (17/025); A61B (17/1671); A61B (2017/0256); A61F (2/4611)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
208227	12/1877	Dorr	N/A	N/A
972983	12/1909	Arthur	N/A	N/A
1328624	12/1919	Graham	N/A	N/A
1548184	12/1924	Cameron	N/A	N/A
2704064	12/1954	Fizzell et al.	N/A	N/A
2736002	12/1955	Oriel	N/A	N/A
2808826	12/1956	Reiner et al.	N/A	N/A
3364929	12/1967	Ide et al.	N/A	N/A
3486505	12/1968	Morrison	N/A	N/A
3518993	12/1969	Blake	N/A	N/A
3604487	12/1970	Gilbert	N/A	N/A
3664329	12/1971	Naylor	N/A	N/A
3682162	12/1971	Colyer	N/A	N/A
3745995	12/1972	Kraus	N/A	N/A
3785368	12/1973	McCarthy et al.	N/A	N/A

3830226	12/1973	Staub et al.	N/A	N/A
3848601	12/1973	Ma et al.	N/A	N/A
3867728	12/1974	Stubstad	N/A	N/A
3957036	12/1975	Normann	N/A	N/A
4026304	12/1976	Levy	N/A	N/A
4026305	12/1976	Brownlee et al.	N/A	N/A
4099519	12/1977	Warren	N/A	N/A
4164214	12/1978	Stark et al.	N/A	N/A
4207897	12/1979	Lloyd et al.	N/A	N/A
4224949	12/1979	Scott et al.	N/A	N/A
4226228	12/1979	Shin et al.	N/A	N/A
4235242	12/1979	Howson et al.	N/A	N/A
4285347	12/1980	Hess	N/A	N/A
4291705	12/1980	Severinghaus et al.	N/A	N/A
4461300	12/1983	Christensen	N/A	N/A
4501269	12/1984	Bagby	N/A	N/A
4515168	12/1984	Chester et al.	N/A	N/A
4519403	12/1984	Dickhudt	N/A	N/A
4545374	12/1984	Jacobson	604/164.11	A61B 17/32056
4561445	12/1984	Berke et al.	N/A	N/A
4562832	12/1985	Wilder et al.	N/A	N/A
4573448	12/1985	Kambin	N/A	N/A
4592369	12/1985	Davis et al.	N/A	N/A
4595018	12/1985	Rantala	N/A	N/A
4611597	12/1985	Kraus	N/A	N/A
4633889	12/1986	Talalla	N/A	N/A
4646738	12/1986	Trott	N/A	N/A
4657550	12/1986	Daher	N/A	N/A
4658835	12/1986	Pohndorf	N/A	N/A
4743256	12/1987	Brantigan	N/A	N/A
4744371	12/1987	Harris	N/A	N/A
4759377	12/1987	Dykstra	N/A	N/A
4781591	12/1987	Allen	N/A	N/A
4784150	12/1987	Voorhies et al.	N/A	N/A
4807642	12/1988	Brown	N/A	N/A
4834757	12/1988	Brantigan	N/A	N/A
4877020	12/1988	Vich	606/86R	A61F 2/4601
4878915	12/1988	Brantigan	N/A	N/A
4892105	12/1989	Prass	N/A	N/A
4913134	12/1989	Luque	N/A	N/A
4926865	12/1989	Oman	N/A	N/A
4932975	12/1989	Main et al.	N/A	N/A
4950296	12/1989	McIntyre	N/A	N/A
4961740	12/1989	Ray et al.	N/A	N/A
4962766	12/1989	Herzon	N/A	N/A
4964411	12/1989	Johnson et al.	N/A	N/A
5007902	12/1990	Witt	N/A	N/A
5015247	12/1990	Michelson	N/A	N/A

5026373	12/1990	Ray et al.	N/A	N/A
5047055	12/1990	Bao et al.	N/A	N/A
5052373	12/1990	Michelson	N/A	N/A
5055104	12/1990	Ray	N/A	N/A
5058602	12/1990	Brody	N/A	N/A
5062845	12/1990	Kuslich et al.	N/A	N/A
5071437	12/1990	Steffee	N/A	N/A
5081990	12/1991	Deletis	N/A	N/A
5092344	12/1991	Lee	N/A	N/A
5092572	12/1991	Litwak et al.	N/A	N/A
5127403	12/1991	Brownlee	N/A	N/A
5133717	12/1991	Chopin	N/A	N/A
5133755	12/1991	Brekke	N/A	N/A
5161533	12/1991	Prass et al.	N/A	N/A
5171278	12/1991	Pisharodi	N/A	N/A
5171279	12/1991	Mathews	N/A	N/A
5192327	12/1992	Brantigan	N/A	N/A
5195541	12/1992	Obenchain	N/A	N/A
5196015	12/1992	Neubardt	N/A	N/A
5217497	12/1992	Mehdian	N/A	N/A
RE34390	12/1992	Culver	N/A	N/A
5255691	12/1992	Otten	N/A	N/A
5263953	12/1992	Bagby	N/A	N/A
5269785	12/1992	Bonutti	N/A	N/A
5282468	12/1993	Klepinski	N/A	N/A
5284153	12/1993	Raymond et al.	N/A	N/A
5284154	12/1993	Raymond et al.	N/A	N/A
5290494	12/1993	Coombes et al.	N/A	N/A
5295994	12/1993	Bonutti	N/A	N/A
5299563	12/1993	Seton	N/A	N/A
5300076	12/1993	Lerich	N/A	N/A
5304210	12/1993	Crook	N/A	N/A
5306307	12/1993	Senter et al.	N/A	N/A
5306309	12/1993	Wagner et al.	N/A	N/A
5312417	12/1993	Wilk	N/A	N/A
5313956	12/1993	Knutsson et al.	N/A	N/A
5313962	12/1993	Obenchain	N/A	N/A
5322505	12/1993	Krause et al.	N/A	N/A
5327902	12/1993	Lemmen	N/A	N/A
5331975	12/1993	Bonutti	N/A	N/A
5333618	12/1993	Lekhtman et al.	N/A	N/A
5334205	12/1993	Cain	N/A	N/A
5336223	12/1993	Rogers	N/A	N/A
5342384	12/1993	Sugarbaker	N/A	N/A
5364400	12/1993	Rego, Jr. et al.	N/A	N/A
5375067	12/1993	Berchin	N/A	N/A
5375594	12/1993	Cueva	N/A	N/A
5383876	12/1994	Nardella	N/A	N/A
5395317	12/1994	Kambin	N/A	N/A
5395372	12/1994	Holt et al.	N/A	N/A

5397363	12/1994	Gelbard	N/A	N/A
5397364	12/1994	Kozak	N/A	N/A
5405391	12/1994	Henderson et al.	N/A	N/A
5413602	12/1994	Metz- Stavenhagen	N/A	N/A
5425772	12/1994	Brantigan	N/A	N/A
5431658	12/1994	Moskovich	N/A	N/A
5443514	12/1994	Steffee	N/A	N/A
5443515	12/1994	Cohen et al.	N/A	N/A
5445639	12/1994	Kuslich et al.	N/A	N/A
5450845	12/1994	Alexgaard	N/A	N/A
5454811	12/1994	Huebner	N/A	N/A
5458638	12/1994	Kuslich et al.	N/A	N/A
5458642	12/1994	Beer et al.	N/A	N/A
5474057	12/1994	Makower et al.	N/A	N/A
5474558	12/1994	Neubardt	N/A	N/A
5480440	12/1995	Kambin	N/A	N/A
5482038	12/1995	Ruff	N/A	N/A
5484403	12/1995	Yoakum et al.	N/A	N/A
5484437	12/1995	Michelson	N/A	N/A
5489307	12/1995	Kuslich et al.	N/A	N/A
5489308	12/1995	Kuslich et al.	N/A	N/A
5509893	12/1995	Pracas	N/A	N/A
5514153	12/1995	Bonutti	N/A	N/A
5514180	12/1995	Heggeness et al.	N/A	N/A
5522879	12/1995	Scopelianos	N/A	N/A
5522899	12/1995	Michelson	N/A	N/A
5524624	12/1995	Tepper et al.	N/A	N/A
5527312	12/1995	Ray	N/A	N/A
5534030	12/1995	Navarro et al.	N/A	N/A
5540235	12/1995	Wilson	N/A	N/A
5540688	12/1995	Navas	N/A	N/A
5545222	12/1995	Bonutti	N/A	N/A
5549656	12/1995	Reiss	N/A	N/A
5560372	12/1995	Cory	N/A	N/A
5562736	12/1995	Ray et al.	N/A	N/A
5565005	12/1995	Erickson et al.	N/A	N/A
5566678	12/1995	Cadwell	N/A	N/A
5571149	12/1995	Liss et al.	N/A	N/A
5571190	12/1995	Ulrich	N/A	N/A
5571192	12/1995	Schonhoffer	N/A	N/A
5579781	12/1995	Cooke	N/A	N/A
5593409	12/1996	Michelson	N/A	N/A
5593429	12/1996	Ruff	N/A	N/A
5599279	12/1996	Slotman et al.	N/A	N/A
5607424	12/1996	Tropiano	N/A	N/A
5609636	12/1996	Kohrs et al.	N/A	N/A
5611800	12/1996	Davis et al.	N/A	N/A
5611810	12/1996	Arnold et al.	N/A	N/A
5630813	12/1996	Kieturakis	N/A	N/A

5632747	12/1996	Scarborough et al.	N/A	N/A
5645598	12/1996	Brosnahan et al.	N/A	N/A
5653761	12/1996	Pisharodi	N/A	N/A
5653762	12/1996	Pisharodi	N/A	N/A
5658336	12/1996	Pisdharodi	N/A	N/A
5658337	12/1996	Kohrs et al.	N/A	N/A
5662710	12/1996	Bonutti	N/A	N/A
5665122	12/1996	Kambin	N/A	N/A
5667508	12/1996	Errico et al.	N/A	N/A
5669909	12/1996	Zdeblick et al.	N/A	N/A
5671752	12/1996	Sinderby et al.	N/A	N/A
5676703	12/1996	Gelbard	N/A	N/A
5681265	12/1996	Maeda et al.	N/A	N/A
5683394	12/1996	Rinner	N/A	N/A
5683400	12/1996	McGuire	N/A	N/A
5683464	12/1996	Wagner et al.	N/A	N/A
5690629	12/1996	Asher et al.	N/A	N/A
5700264	12/1996	Zucherman et al.	N/A	N/A
5700291	12/1996	Kuslich et al.	N/A	N/A
5700292	12/1996	Marguiles	N/A	N/A
5702449	12/1996	McKay	N/A	N/A
5702451	12/1996	Biedermann et al.	N/A	N/A
5702453	12/1996	Rabbe et al.	N/A	N/A
5702454	12/1996	Baumgartner	N/A	N/A
5702455	12/1996	Saggar	N/A	N/A
5703451	12/1996	Yamamichi et al.	N/A	N/A
5707359	12/1997	Bufalini	N/A	N/A
5707373	12/1997	Sevrain et al.	N/A	N/A
5711307	12/1997	Smits	N/A	N/A
5711957	12/1997	Patat et al.	N/A	N/A
5716415	12/1997	Steffee	N/A	N/A
5720748	12/1997	Kuslich et al.	N/A	N/A
5720751	12/1997	Jackson	N/A	N/A
5728046	12/1997	Mayer et al.	N/A	N/A
5728159	12/1997	Stroeve et al.	N/A	N/A
5741253	12/1997	Michelson	N/A	N/A
5741261	12/1997	Moskovitz et al.	N/A	N/A
5755797	12/1997	Baumgartner	N/A	N/A
5759159	12/1997	Masreliez	N/A	N/A
5762629	12/1997	Kambin	N/A	N/A
5766252	12/1997	Henry et al.	N/A	N/A
5772661	12/1997	Michelson	N/A	N/A
5775331	12/1997	Raymond et al.	N/A	N/A
5775797	12/1997	Henstra	N/A	N/A
5776144	12/1997	Leysieffer et al.	N/A	N/A
5779642	12/1997	Nightengale	N/A	N/A
5782830	12/1997	Farris	N/A	N/A
5782919	12/1997	Zdeblick et al.	N/A	N/A
5785658	12/1997	Benaron	N/A	N/A
5785710	12/1997	Michelson	N/A	N/A

5792044	12/1997	Foley et al.	N/A	N/A
5797854	12/1997	Hedgecock	N/A	N/A
5797909	12/1997	Michelson	N/A	N/A
5800549	12/1997	Bao et al.	N/A	N/A
5800550	12/1997	Sertich	N/A	N/A
5814073	12/1997	Bonutti	N/A	N/A
5814084	12/1997	Grivas et al.	N/A	N/A
5830151	12/1997	Hadzic et al.	N/A	N/A
5851191	12/1997	Gozani	N/A	N/A
5851208	12/1997	Trott	N/A	N/A
5853373	12/1997	Griffith et al.	N/A	N/A
5860973	12/1998	Michelson	N/A	N/A
5862314	12/1998	Jeddeloh	N/A	N/A
5865845	12/1998	Thalgott	N/A	N/A
5865848	12/1998	Baker	N/A	N/A
5872314	12/1998	Clinton	N/A	N/A
5885210	12/1998	Cox	N/A	N/A
5885219	12/1998	Nightengale	N/A	N/A
5885299	12/1998	Winslow et al.	N/A	N/A
5888196	12/1998	Bonutti	N/A	N/A
5888219	12/1998	Bonutti	N/A	N/A
5888224	12/1998	Beckers et al.	N/A	N/A
5891147	12/1998	Moskovitz et al.	N/A	N/A
5893890	12/1998	Pisharodi	N/A	N/A
5902231	12/1998	Foley et al.	N/A	N/A
5904719	12/1998	Errico et al.	N/A	N/A
5910315	12/1998	Stevenson et al.	N/A	N/A
5928139	12/1998	Korns et al.	N/A	N/A
5928158	12/1998	Aristides	N/A	N/A
5931777	12/1998	Sava	N/A	N/A
5935131	12/1998	Bonutti et al.	N/A	N/A
5938688	12/1998	Schiff	N/A	N/A
5942698	12/1998	Stevens	N/A	N/A
5944658	12/1998	Korns et al.	N/A	N/A
5954769	12/1998	Rosenlicht	N/A	N/A
5968098	12/1998	Winslow	N/A	N/A
5976094	12/1998	Gozani et al.	N/A	N/A
5993474	12/1998	Ouchi	N/A	N/A
6003426	12/1998	Kobayashi et al.	N/A	N/A
6004262	12/1998	Putz et al.	N/A	N/A
6004312	12/1998	Finneran	N/A	N/A
6004326	12/1998	Castro et al.	N/A	N/A
6007487	12/1998	Foley et al.	N/A	N/A
6008433	12/1998	Stone	N/A	N/A
6015436	12/1999	Schunhuffer	N/A	N/A
6024696	12/1999	Hoftman et al.	N/A	N/A
6027456	12/1999	Feler et al.	N/A	N/A
6033405	12/1999	Winslow et al.	N/A	N/A
6038469	12/1999	Karlsson et al.	N/A	N/A
6038477	12/1999	Kayyali	N/A	N/A

6039761	12/1999	Li et al.	N/A	N/A
6042582	12/1999	Ray	N/A	N/A
6045580	12/1999	Scarborough et al.	N/A	N/A
6048342	12/1999	Zucherman et al.	N/A	N/A
6050992	12/1999	Nichols	N/A	N/A
6059829	12/1999	Schlapfer et al.	N/A	N/A
6063088	12/1999	Winslow	N/A	N/A
6066175	12/1999	Henderson	N/A	N/A
6074343	12/1999	Nathanson et al.	N/A	N/A
6083154	12/1999	Liu et al.	N/A	N/A
6083225	12/1999	Winslow et al.	N/A	N/A
6095987	12/1999	Schmulewitz	N/A	N/A
6096080	12/1999	Nicholson et al.	N/A	N/A
6102948	12/1999	Brosnahan, III	N/A	N/A
6104957	12/1999	Alo et al.	N/A	N/A
6104960	12/1999	Duysens et al.	N/A	N/A
6120503	12/1999	Michelson	N/A	N/A
6120506	12/1999	Kohrs et al.	N/A	N/A
6126660	12/1999	Dietz	N/A	N/A
6132386	12/1999	Gozani et al.	N/A	N/A
6132387	12/1999	Gozani et al.	N/A	N/A
6132472	12/1999	Bonutti	N/A	N/A
6135965	12/1999	Turner et al.	N/A	N/A
6139493	12/1999	Korns et al.	N/A	N/A
6143033	12/1999	Paul et al.	N/A	N/A
6146335	12/1999	Gozani	N/A	N/A
6152871	12/1999	Foley et al.	N/A	N/A
6159211	12/1999	Boriani et al.	N/A	N/A
6159215	12/1999	Urbahns et al.	N/A	N/A
6161047	12/1999	King et al.	N/A	N/A
6174311	12/2000	Branch et al.	N/A	N/A
6181961	12/2000	Prass	N/A	N/A
6193756	12/2000	Studer et al.	N/A	N/A
6196969	12/2000	Bester et al.	N/A	N/A
6200347	12/2000	Anderson	N/A	N/A
6206826	12/2000	Mathews et al.	N/A	N/A
6217509	12/2000	Foley et al.	N/A	N/A
6224549	12/2000	Drongelen	N/A	N/A
6224607	12/2000	Michelson	N/A	N/A
6224631	12/2000	Kohrs	N/A	N/A
6235059	12/2000	Benezech et al.	N/A	N/A
6241769	12/2000	Nicholson et al.	N/A	N/A
6241771	12/2000	Gresser et al.	N/A	N/A
6245108	12/2000	Biscup	N/A	N/A
6251140	12/2000	Marino et al.	N/A	N/A
6258125	12/2000	Paul et al.	N/A	N/A
6259945	12/2000	Epstein et al.	N/A	N/A
6264651	12/2000	Underwood et al.	N/A	N/A
6266558	12/2000	Gozani et al.	N/A	N/A
6273905	12/2000	Streeter	N/A	N/A

6277149	12/2000	Boyle et al.	N/A	N/A
6292701	12/2000	Prass et al.	N/A	N/A
6306100	12/2000	Prass	N/A	N/A
6308712	12/2000	Shaw	N/A	N/A
6312392	12/2000	Herzon	N/A	N/A
6319257	12/2000	Carignan et al.	N/A	N/A
6325764	12/2000	Griffith et al.	N/A	N/A
6334068	12/2000	Hacker	N/A	N/A
6348058	12/2001	Melkent et al.	N/A	N/A
6360750	12/2001	Gerber et al.	N/A	N/A
6371968	12/2001	Kogasaka et al.	N/A	N/A
6371989	12/2001	Chauvin et al.	N/A	N/A
6383221	12/2001	Scarborough et al.	N/A	N/A
6409766	12/2001	Brett	N/A	N/A
6425772	12/2001	Bernier et al.	N/A	N/A
6425859	12/2001	Foley et al.	N/A	N/A
6425901	12/2001	Zhu et al.	N/A	N/A
6432140	12/2001	Lin	N/A	N/A
6440142	12/2001	Ralph et al.	N/A	N/A
6442814	12/2001	Landry et al.	N/A	N/A
6447547	12/2001	Michelson	N/A	N/A
6450952	12/2001	Rioux et al.	N/A	N/A
6451015	12/2001	Rittman, III et al.	N/A	N/A
6454806	12/2001	Cohen et al.	N/A	N/A
6466817	12/2001	Kaula et al.	N/A	N/A
6468205	12/2001	Mollenauer et al.	N/A	N/A
6468207	12/2001	Fowler, Jr.	N/A	N/A
6468311	12/2001	Boyd et al.	N/A	N/A
6491724	12/2001	Ferree	N/A	N/A
6500128	12/2001	Marino	N/A	N/A
6520907	12/2002	Foley et al.	N/A	N/A
6524320	12/2002	DiPoto	N/A	N/A
6527773	12/2002	Lin et al.	N/A	N/A
6535759	12/2002	Epstein et al.	N/A	N/A
D472634	12/2002	Anderson	N/A	N/A
D473650	12/2002	Anderson	N/A	N/A
6547823	12/2002	Scarborough et al.	N/A	N/A
6558423	12/2002	Michelson	N/A	N/A
6564078	12/2002	Marino et al.	N/A	N/A
6576017	12/2002	Foley et al.	N/A	N/A
6579244	12/2002	Goodwin	N/A	N/A
6595998	12/2002	Johnson et al.	N/A	N/A
6620157	12/2002	Dabney et al.	N/A	N/A
6626905	12/2002	Schmiel et al.	N/A	N/A
6635086	12/2002	Lin	N/A	N/A
6645194	12/2002	Briscoe et al.	N/A	N/A
6648895	12/2002	Burkus et al.	N/A	N/A
6672019	12/2003	Wenz	N/A	N/A
6676703	12/2003	Biscup	N/A	N/A
6679833	12/2003	Smith et al.	N/A	N/A

6706067	12/2003	Shimp et al.	N/A	N/A
6719692	12/2003	Kleffner et al.	N/A	N/A
6743255	12/2003	Ferree	N/A	N/A
6746484	12/2003	Liu et al.	N/A	N/A
6755841	12/2003	Fraser et al.	N/A	N/A
6760616	12/2003	Hoey et al.	N/A	N/A
6761739	12/2003	Shepard	N/A	N/A
6770074	12/2003	Michelson	N/A	N/A
6796985	12/2003	Bolger et al.	N/A	N/A
6810281	12/2003	Brock et al.	N/A	N/A
6819956	12/2003	DiLorenzo	N/A	N/A
6824564	12/2003	Crozet	N/A	N/A
6847849	12/2004	Mamo et al.	N/A	N/A
6849047	12/2004	Goodwin	N/A	N/A
6855105	12/2004	Jackson, III et al.	N/A	N/A
6871099	12/2004	Whitehurst et al.	N/A	N/A
D503801	12/2004	Jackson	N/A	N/A
6902569	12/2004	Parmer et al.	N/A	N/A
6923811	12/2004	Carl et al.	N/A	N/A
6923814	12/2004	Hildebrand et al.	N/A	N/A
6926728	12/2004	Zucherman et al.	N/A	N/A
6929606	12/2004	Ritland	N/A	N/A
6942697	12/2004	Lange et al.	N/A	N/A
6942698	12/2004	Jackson	N/A	N/A
6945933	12/2004	Branch	N/A	N/A
6951538	12/2004	Ritland	N/A	N/A
6964687	12/2004	Bernard et al.	N/A	N/A
6979353	12/2004	Bresina	N/A	N/A
6984245	12/2005	McGahan et al.	N/A	N/A
6986788	12/2005	Paul et al.	N/A	N/A
6989031	12/2005	Michelson	N/A	N/A
7018416	12/2005	Hanson et al.	N/A	N/A
7033394	12/2005	Michelson	N/A	N/A
7047082	12/2005	Schrom et al.	N/A	N/A
7050848	12/2005	Hoey et al.	N/A	N/A
7079883	12/2005	Marino et al.	N/A	N/A
7089059	12/2005	Pless	N/A	N/A
D530423	12/2005	Miles et al.	N/A	N/A
7177677	12/2006	Kaula et al.	N/A	N/A
7198598	12/2006	Smith et al.	N/A	N/A
7207949	12/2006	Miles et al.	N/A	N/A
7226451	12/2006	Shluzas et al.	N/A	N/A
7226480	12/2006	Thalgott	N/A	N/A
7261688	12/2006	Smith et al.	N/A	N/A
7470236	12/2007	Kelleher et al.	N/A	N/A
7473222	12/2008	Dewey et al.	N/A	N/A
7481766	12/2008	Lee et al.	N/A	N/A
7491237	12/2008	Randall et al.	N/A	N/A
7522953	12/2008	Kaula et al.	N/A	N/A
7556601	12/2008	Branch et al.	N/A	N/A

7582058	12/2008	Miles et al.	N/A	N/A
7643884	12/2009	Pond et al.	N/A	N/A
7691057	12/2009	Miles et al.	N/A	N/A
7717959	12/2009	William et al.	N/A	N/A
D620110	12/2009	Courtney et al.	N/A	N/A
7763078	12/2009	Peterman et al.	N/A	N/A
7815682	12/2009	Peterson et al.	N/A	N/A
7819801	12/2009	Miles et al.	N/A	N/A
7935051	12/2010	Miles et al.	N/A	N/A
8043377	12/2010	Guyer et al.	N/A	N/A
8083799	12/2010	Baynham et al.	N/A	N/A
D653756	12/2011	Courtney et al.	N/A	N/A
8216312	12/2011	Gray	N/A	N/A
8287597	12/2011	Pimenta	623/17.16	A61B 17/320016
8313430	12/2011	Pimenta	N/A	N/A
8328856	12/2011	Donahoe et al.	N/A	N/A
8623088	12/2013	Tohmeh et al.	N/A	N/A
9314152	12/2015	Pimenta	N/A	N/A
9351845	12/2015	Pimenta	N/A	N/A
9795367	12/2016	Lee	N/A	N/A
10426627	12/2018	Pimenta	N/A	A61F 2/4455
2001/0039949	12/2000	Loubser	N/A	N/A
2001/0053914	12/2000	Landry	606/105	A61F 2/4611
2001/0056280	12/2000	Underwood et al.	N/A	N/A
2002/0007129	12/2001	Marino	N/A	N/A
2002/0010392	12/2001	Desai	N/A	N/A
2002/0058950	12/2001	Winterbottom et al.	N/A	N/A
2002/0072686	12/2001	Hoey et al.	N/A	N/A
2002/0077632	12/2001	Tsou	N/A	N/A
2002/0123780	12/2001	Grill et al.	N/A	N/A
2002/0138147	12/2001	Cohen	N/A	N/A
2002/0161415	12/2001	Cohen et al.	N/A	N/A
2002/0193843	12/2001	Hill et al.	N/A	N/A
2003/0032966	12/2002	Foley et al.	N/A	N/A
2003/0100950	12/2002	Moret	N/A	N/A
2003/0105503	12/2002	Marino	N/A	N/A
2003/0105528	12/2002	Shimp et al.	N/A	N/A
2003/0109928	12/2002	Pasquet et al.	N/A	N/A
2003/0125739	12/2002	Bagga et al.	N/A	N/A
2003/0139648	12/2002	Foley et al.	N/A	N/A
2003/0139812	12/2002	Garcia et al.	N/A	N/A
2003/0149341	12/2002	Clifton	N/A	N/A
2003/0153975	12/2002	Byrd et al.	N/A	N/A
2003/0181981	12/2002	Lemaire	N/A	N/A
2003/0225405	12/2002	Weiner	N/A	N/A
2003/0236544	12/2002	Lunsford et al.	N/A	N/A
2004/0082999	12/2003	Mathys et al.	N/A	N/A
2004/0117018	12/2003	Michelson	N/A	N/A

2004/0127990	12/2003	Bartish et al.	N/A	N/A
2004/0153155	12/2003	Chung et al.	N/A	N/A
2004/0158324	12/2003	Lange	N/A	N/A
2004/0199084	12/2003	Kelleher et al.	N/A	N/A
2004/0225228	12/2003	Ferree	N/A	N/A
2005/0004593	12/2004	Simonson	N/A	N/A
2005/0004623	12/2004	Miles et al.	N/A	N/A
2005/0010292	12/2004	Carrasco	N/A	N/A
2005/0033380	12/2004	Tanner et al.	N/A	N/A
2005/0075578	12/2004	Gharib et al.	N/A	N/A
2005/0080320	12/2004	Lee et al.	N/A	N/A
2005/0090829	12/2004	Martz	N/A	N/A
2005/0119753	12/2004	McGahan et al.	N/A	N/A
2005/0143822	12/2004	Paul	N/A	N/A
2005/0149035	12/2004	Pimenta	606/86R	A61B 1/32
2005/0177245	12/2004	Leatherbury et al.	N/A	N/A
2005/0182454	12/2004	Gharib et al.	N/A	N/A
2005/0187629	12/2004	Michelson	N/A	N/A
2005/0192575	12/2004	Pacheco	N/A	N/A
2005/0197702	12/2004	Coppes et al.	N/A	N/A
2005/0216083	12/2004	Michelson	N/A	N/A
2006/0025703	12/2005	Miles et al.	N/A	N/A
2006/0041258	12/2005	Galea	N/A	N/A
2006/0052828	12/2005	Kim et al.	N/A	N/A
2006/0069315	12/2005	Miles et al.	N/A	N/A
2006/0074426	12/2005	Lieberman	606/79	A61B 17/1604
2006/0116766	12/2005	Lemaire	N/A	N/A
2006/0129244	12/2005	Ensign	N/A	N/A
2006/0149267	12/2005	Nordt	606/79	A61B 17/320016
2006/0224078	12/2005	Hoey et al.	N/A	N/A
2006/0235520	12/2005	Pannu	N/A	N/A
2007/0016097	12/2006	Farquhar et al.	N/A	N/A
2007/0100212	12/2006	Pimenta	N/A	N/A
2007/0191945	12/2006	Yu et al.	N/A	N/A
2007/0198062	12/2006	Miles et al.	N/A	N/A
2007/0213826	12/2006	Smith et al.	N/A	N/A
2007/0233253	12/2006	Bray et al.	N/A	N/A
2007/0270873	12/2006	Flickinger	606/90	A61F 2/4611
2007/0270972	12/2006	Gordon et al.	N/A	N/A
2007/0293782	12/2006	Marino	N/A	N/A
2008/0058606	12/2007	Miles et al.	N/A	N/A
2008/0064976	12/2007	Kelleher et al.	N/A	N/A
2008/0064977	12/2007	Kelleher et al.	N/A	N/A
2008/0065178	12/2007	Kelleher et al.	N/A	N/A
2008/0065219	12/2007	Dye	N/A	N/A
2008/0071191	12/2007	Kelleher et al.	N/A	N/A
2008/0077247	12/2007	Murillo et al.	N/A	N/A
2008/0097164	12/2007	Miles et al.	N/A	N/A
2008/0119932	12/2007	Lechmann	N/A	N/A

2008/0147194	12/2007	Grotz et al.	N/A	N/A
2008/0154375	12/2007	Serhan	N/A	N/A
2008/0255563	12/2007	Farr	N/A	N/A
2008/0294262	12/2007	Levieux	N/A	N/A
2008/0300465	12/2007	Feigenwinter et al.	N/A	N/A
2008/0312742	12/2007	Abernathie	N/A	N/A
2009/0054987	12/2008	Chin et al.	N/A	N/A
2009/0062917	12/2008	Foley et al.	N/A	N/A
2009/0124860	12/2008	Miles et al.	N/A	N/A
2009/0138050	12/2008	Ferree	N/A	N/A
2009/0182430	12/2008	Tyber et al.	N/A	N/A
2009/0192403	12/2008	Gharib et al.	N/A	N/A
2009/0192613	12/2008	Wing et al.	N/A	N/A
2009/0198339	12/2008	Kleiner et al.	N/A	N/A
2009/0204016	12/2008	Gharib et al.	N/A	N/A
2009/0210062	12/2008	Thalgott et al.	N/A	N/A
2009/0254182	12/2008	Kovarik et al.	N/A	N/A
2009/0299479	12/2008	Jones et al.	N/A	N/A
2010/0069783	12/2009	Miles et al.	N/A	N/A
2010/0130827	12/2009	Pimenta et al.	N/A	N/A
2010/0152603	12/2009	Miles et al.	N/A	N/A
2010/0152853	12/2009	Kirschman	N/A	N/A
2010/0160738	12/2009	Miles et al.	N/A	N/A
2010/0161057	12/2009	Berry et al.	N/A	N/A
2010/0174148	12/2009	Miles et al.	N/A	N/A
2010/0228296	12/2009	Vraney et al.	N/A	N/A
2010/0262246	12/2009	Attia	N/A	N/A
2011/0040382	12/2010	Muhanna	N/A	N/A
2011/0046736	12/2010	Graf	N/A	N/A
2011/0125266	12/2010	Rodgers	N/A	N/A
2011/0166657	12/2010	Thalgott	N/A	N/A
2011/0196493	12/2010	Pimenta	N/A	N/A
2011/0208309	12/2010	Peterson et al.	N/A	N/A
2011/0230970	12/2010	Lynn et al.	N/A	N/A
2011/0245923	12/2010	Cobb et al.	N/A	N/A
2011/0301710	12/2010	Mather et al.	N/A	N/A
2011/0313528	12/2010	Laubert et al.	N/A	N/A
2011/0319943	12/2010	Donahoe et al.	N/A	N/A
2011/0320000	12/2010	O'Neil et al.	N/A	N/A
2012/0078370	12/2011	James et al.	N/A	N/A
2012/0089228	12/2011	Poulos	N/A	N/A
2012/0232664	12/2011	Ulrich, Jr. et al.	N/A	N/A
2012/0245690	12/2011	Cowan, Jr. et al.	N/A	N/A
2012/0277866	12/2011	Kalluri et al.	N/A	N/A
2012/0277867	12/2011	Kana et al.	N/A	N/A
2012/0283521	12/2011	Smith	N/A	N/A
2012/0303128	12/2011	Ullrich, Jr. et al.	N/A	N/A
2012/0310351	12/2011	Farley	N/A	N/A
2012/0316649	12/2011	Johnston et al.	N/A	N/A

2013/0018466	12/2012	Yu et al.	N/A	N/A
2013/0018471	12/2012	Davenport et al.	N/A	N/A
2013/0023992	12/2012	Moskowitz et al.	N/A	N/A
2013/0053967	12/2012	Sournac et al.	N/A	N/A
2013/0060336	12/2012	Hooper et al.	N/A	N/A
2013/0060339	12/2012	Duffield et al.	N/A	N/A
2013/0066428	12/2012	Biedermann et al.	N/A	N/A
2013/0073044	12/2012	Gamache	N/A	N/A
2013/0073047	12/2012	Laskowitz et al.	N/A	N/A
2013/0096683	12/2012	Kube, II	N/A	N/A
2013/0116790	12/2012	Seifert	N/A	N/A
2013/0123923	12/2012	Pavlov et al.	N/A	N/A
2013/0131812	12/2012	Ganey	N/A	N/A
2013/0138214	12/2012	Greenhalgh et al.	N/A	N/A
2013/0158663	12/2012	Miller et al.	N/A	N/A
2013/0158664	12/2012	Palmatier et al.	N/A	N/A
2013/0158669	12/2012	Sungarian et al.	N/A	N/A
2013/0166027	12/2012	Bellas	N/A	N/A
2013/0190875	12/2012	Shulock et al.	N/A	N/A
2013/0197642	12/2012	Ernst	N/A	N/A
2013/0238095	12/2012	Pavento	N/A	N/A
2013/0268080	12/2012	Melkent et al.	N/A	N/A
2013/0274883	12/2012	McLuen et al.	N/A	N/A
2013/0297029	12/2012	Kana et al.	N/A	N/A
2014/0012380	12/2013	Laurence	N/A	N/A
2014/0025168	12/2013	Klimek et al.	N/A	N/A
2014/0121777	12/2013	Rosen et al.	N/A	N/A
2014/0128977	12/2013	Glerum et al.	N/A	N/A
2014/0277495	12/2013	Muhanna	N/A	N/A
2014/0379085	12/2013	Duffield et al.	N/A	N/A
2015/0018954	12/2014	Loebl et al.	N/A	N/A
2015/0045892	12/2014	Lynn et al.	N/A	N/A
2015/0088256	12/2014	Ballard	N/A	N/A
2015/0100124	12/2014	Whipple	N/A	N/A
2018/0214279	12/2017	Etminan	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2015507	12/1998	CA	N/A
299 08 259	12/1998	DE	N/A
369603	12/1989	EP	N/A
517030	12/1991	EP	N/A
667127	12/1994	EP	N/A
706876	12/1995	EP	N/A
716840	12/1995	EP	N/A
737448	12/1995	EP	N/A
796593	12/1996	EP	N/A
880938	12/1997	EP	N/A
809974	12/1997	EP	N/A

809975	12/1997	EP	N/A
811356	12/1997	EP	N/A
0 972 538	12/1999	EP	N/A
2 795 624	12/2000	FR	N/A
90/00037	12/1989	WO	N/A
91/06261	12/1990	WO	N/A
92/14423	12/1991	WO	N/A
94/04100	12/1993	WO	N/A
94/10928	12/1993	WO	N/A
95/01810	12/1994	WO	N/A
96/08205	12/1995	WO	N/A
96/17564	12/1995	WO	N/A
96/41582	12/1995	WO	N/A
97/20513	12/1996	WO	N/A
97/33525	12/1996	WO	N/A
97/37620	12/1996	WO	N/A
98/09586	12/1997	WO	N/A
98/14142	12/1997	WO	N/A
98/17208	12/1997	WO	N/A
98/25539	12/1997	WO	N/A
99/08627	12/1998	WO	N/A
99/38461	12/1998	WO	N/A
00/38574	12/1999	WO	N/A
00/45712	12/1999	WO	N/A
00/45713	12/1999	WO	N/A
00/66217	12/1999	WO	N/A
00/67645	12/1999	WO	N/A
01/37728	12/2000	WO	N/A
01/41681	12/2000	WO	N/A
01/49333	12/2000	WO	N/A
02/054960	12/2001	WO	N/A
03/005887	12/2002	WO	N/A
03/026482	12/2002	WO	N/A
03/037170	12/2002	WO	N/A
05/013805	12/2004	WO	N/A
05/030318	12/2004	WO	N/A
06/042241	12/2005	WO	N/A
06/066217	12/2005	WO	N/A

OTHER PUBLICATIONS

Anatomy of the Lumbar Spine in MED TM MicroEndoscopic Discectomy (1997 Ludann Grand Rapids MI), 14 pgs. cited by applicant

Dirksmeier et al., "Microendoscopic and Open Laminotomy and Discectomy in Lumbar Disc Disease" Seminars in Spine Surgery, 1999, 11(2): 138-146. cited by applicant

METRx Delivered Order Form, 1999, 13 pages. cited by applicant

Medtronic Sofamor Danek "METRx™ MicroDiscectomy System," Medtronic Sofamor Danek USA. 2000, 21 pgs. cited by applicant

Medtronic Sofamor Danek "METRx System Surgical Technique," 2004, 22 pages. cited by applicant

"MetRx System MicroEndoscopic Discectomy: An Evolution in Minimally Invasive Spine

Surgery,” Sofamor Danek, 1999, 6 pages. cited by applicant

Smith and Foley “MetRx System MicroEndoscopic Discectomy: Surgical Technique” Medtronic Sofamor Danek, 2000, 24 pages. cited by applicant

“Sofamor Danek MED Microendoscopic Discectomy System Brochure” including Rapp “New endoscopic lumbar technique improves access preserves tissue” Reprinted with permission from: Ortho12edics Today, 1998, 18(1): 2 pages. cited by applicant

Japanese Patent Office JP Patent Application No. 2006-528306 Office Action with English Translation, Jun. 10, 2009, 4 pages. cited by applicant

Axon 501(k) Notification: Epoch 2000 Neurological Workstation, Dec. 3, 1997, 464 pages. cited by applicant

Foley and Smith, “Microendoscopic Discectomy,” Techniques in Neurosurgery, 1997, 3(4):301-307. cited by applicant

Medtronic Sofamor Danek “Union™/ Union-L™ Anterior & Lateral Impacted Fusion Devices: Clear choice of stabilization,” Medtronic Sofamor Danek, 2000, 4 pages. cited by applicant

NuVasive Vector™ Cannulae, 2000, 1 page. cited by applicant

NuVasive Triad™ Tri-Columnar Spinal EndoArthrodesis™ via Minimally Invasive Guidance, 2000, 1 page (prior to Sep. 25, 2003). cited by applicant

NuVasive Triad™M Cortical Bone Allograft, 2000, 1 page (prior to Sep. 25, 2003). cited by applicant

NuVasive Vertebral Body Access System, 2000, 1 page. cited by applicant

Marina, “New Technology for Guided Navigation with Real Time Nerve Surveillance for Minimally Invasive Spine Discectomy & Arthrodesis,” S12ineline, 2000, p. 39. cited by applicant

NuVasive “INS-1 Screw Test,” 2001, 10 pages. cited by applicant

NuVasive letter re 510k Neuro Vision JJB System, Oct. 16, 2001, 5 pages. cited by applicant

NuVasive letter re 510k Guided Arthroscopy System, Oct. 5, 1999, 6 pages. cited by applicant

NuVasive letter re 510k INS-1 Intraoperative Nerve Surveillance System, Nov. 13, 2000, 7 pages. cited by applicant

“NuVasive™ Receives Clearance to Market Two Key Elem Minimally Invasive Spine Surgery System,” Nov. 27, 2001, 20 pages. cited by applicant

Schick et al., “Microendoscopic lumbar discectomy versus open surgery: an intraoperative EMG study,” Eur S12ine J, 2002, 11: 20-26. cited by applicant

NuVasive letter re: 510(k) for Neurovision JJB System (Summary), Sep. 25, 2001, 28 pages. cited by applicant

NuVasive letter re: Special 510(k) Premarket Notification: Neurovision JJB System (Device Description), Jul. 3, 2003, 18 pages. cited by applicant

NuVasive letter re: Special 510(k) Premarket Notification: Neurovision JJB System (Device Description), Mar. 1, 2004, 16 pages. cited by applicant

NuVasive letter re: Special 510(k) Premarket Notification: Neurovision JJB System (Device Description), May 26, 2005, 17 pages. cited by applicant

NuVasive letter re: 510(k) Premarket Notification: Neurovision JJB System (Device Description), Jun. 24, 2005, 16 pages. cited by applicant

NuVasive letter re: Special 510(k) Premarket Notification: Neurovision JJB System (Device Description), Sep. 14, 2006, 17 pages. cited by applicant

NuVasive 510(k) Premarket Notification: Neurovision JJB System (Device Description), Aug. 20, 2007, 8 pages. cited by applicant

NuVasive letter re: 510(k) Premarket Notification: Guided Spinal Arthroscopy System (Device Description), Feb. 1, 1999, 40 pages. cited by applicant

NuVasive 510(k) Premarket Notification: Spinal System (Summary), Apr. 12, 2004, 10 pages. cited by applicant

NuVasive 510(k) Summary NIM Monitor, Sep. 4, 1998, 4 pages. cited by applicant

NuVasive correspondence re 510(k) Premarket Notification INS-1 Intraoperative Nerve Surveillance System: Section IV Device Description, pp. 12-51 (prior to Sep. 25 2003). cited by applicant

Isley et al., "Recent Advances in Intraoperative Neuromonitoring of Spinal Cord Function: Pedicle Screw Stimulation Techniques," *American Journal of Electroneurodiagnostic Technology*, June. 1997,37(2): 93-126. cited by applicant

Mathews et al., "Laparoscopic Discectomy with Anterior Lumbar Interbody Fusion," *Spine*, 1995, 20(16): 1797-1802. cited by applicant

Rose et al., "Persistently Electrified Pedicle Stimulation Instruments in Spinal Instrumentation: Techniques and Protocol Development," *Spine*, 1997, 22(3): 334-343. cited by applicant

"Electromyography System," International Search report from International Application No. PCT/US00/32329, Apr. 27, 2001, 9 pages. cited by applicant

"Nerve Proximity and Status Detection System and Method," International Search Report from International Application No. PCT/US01/18606, Oct. 18, 2001, 6 pages. cited by applicant

"Relative Nerve Movement and Status Detection System and Method," International Search Report from International Application No. PCT/US01/18579, Jan. 15, 2002, 6 pages. cited by applicant

"System and Method for Determining Nerve Proximity Direction and Pathology During Surgery," International Search Report from International Application No. PCT/US02/22247, Mar. 27, 2003, 4 pages. cited by applicant

"System and Methods for Determining Nerve Direction to a Surgical Instrument," International Search Report from International Application No. PCT/US03/02056, Aug. 12, 2003, 5 pages. cited by applicant

"Systems and Methods for Performing Percutaneous Pedicle Integrity Assessments," International Search Report from International Application No. PCT/US02/35047, Aug. 11, 2003, 5 pages. cited by applicant

"Systems and Methods for Performing Surgery Procedures and Assessments," International Search Report from International Application No. PCT/US02/30617, Jun. 5, 2003, 4 pages. cited by applicant

Lenke et al., "Triggered Electromyographic Threshold for Accuracy of Pedicle Screw Placement," *Spine*, 1995, 20(4): 1585-1591. cited by applicant

"Brackmann II EMG System," Medical Electronics, 1999, 4 pages. cited by applicant

"Neurovision SE Nerve Locator/Monitor", RLN Systems Inc. Operators Manual, 1999, 22 pages. cited by applicant

"The Brackmann II EMG Monitoring System," Medical Electronics Co. Operator's Manual Version 1.1, 1995, 50 pages. cited by applicant

"The Nicolet Viking IV," Nicolet Biomedical Products, 1999, 6 pages. cited by applicant

Anderson et al., "Pedicle screws with high electrical resistance: a potential source of error with stimulus-evoked EMG," *Spine*, Department of Orthopaedic Surgery University of Virginia, Jul. 15, 2002, 27(14): 1577-1581. cited by applicant

Bose et al., "Neurophysiologic Monitoring of Spinal Nerve Root Function During Instrumented Posterior Lumbar Spine Surgery," *Spine*, 2002, 27(13):1444-1450. cited by applicant

Calancie et al., "Stimulus-Evoked EMG Monitoring During Transpedicular Lumbosacral Spine Instrumentation" *Spine*, 1994, 19(24): 2780-2786. cited by applicant

Clements et al., "Evoked and Spontaneous Electromyography to Evaluate Lumbosacral Pedicle Screw Placement," *Spine*, 1996, 21(5): 600-604. cited by applicant

Danesh-Clough et al. , "The Use of Evoked EMG in Detecting Misplaced Thoracolumbar Pedicle Screws," *Spine*, Orthopaedic Department Dunedin Hospital, Jun. 15, 2001, 26(12): 1313-1316. cited by applicant

Darden et al., "A Comparison of Impedance and Electromyogram Measurements in Detecting the Presence of Pedicle Wall Breakthrough," *Spine*, Charlotte Spine Center North Carolina, Jan. 15,

1998,23(2): 256-262. cited by applicant

Ebraheim et al., "Anatomic Relations Between the Lumbar Pedicle and the Adjacent Neural Structures," *Spine*, Department of Orthopaedic Surgery Medical College of Ohio, Oct. 15, 1997, 22(20): 2338-2341. cited by applicant

Ford et al. "Electrical Characteristics of Peripheral Nerve Stimulators Implications for Nerve Localization," *Regional Anesthesia*, 1984, 9: 73-77. cited by applicant

Glassman et al., "A Prospective Analysis of Intraoperative Electromyographic Monitoring of Pedicle Screw Placement With Computed Tomographic Scan Confirmation," *Spine*, 1995, 20(12): 1375-1379. cited by applicant

Greenblatt et al., "Needle Nerve Stimulator-Locator: Nerve Blocks with a New Instrument for Locating Nerves," *Anesthesia& Analgesia*, 1962, 41(5): 599-602. cited by applicant

Haig, "Point of view," *Spine*, 2002, 27(24): 2819. cited by applicant

Haig et al., "The Relation Among Spinal Geometry on MRI, Paraspinal Electromyographic Abnormalities, and Age in Persons Referred for Electrodiagnostic Testing of Low Back Symptoms," *Spine*, Department of Physical Medicine and Rehabilitation University of Michigan, Sep. 1, 2002, 27(17): 1918-1925. cited by applicant

Holland et al., "Higher Electrical Stimulus Intensities are Required to Activate Chronically Compressed Nerve Roots: Implications for Intraoperative Electromyographic Pedicle Screw Testing," *Spine*, Department of Neurology, Johns Hopkins University School of Medicine, Jan. 15, 1998, 23(2): 224-227. cited by applicant

Holland, "Intraoperative Electromyography During Thoracolumbar Spinal Surgery," *Spine*, 1998, 23(17): 1915-1922. cited by applicant

Journee et al., "System for Intra-Operative Monitoring of the Cortical Integrity of the Pedicle During Pedicle Screw Placement in Low-Back Surgery: Design and Clinical Results," *Sensory and Neuromuscular Diagnostic Instrumentation and Data Analysis I*, 18th Annual International Conference on Engineering in Medicine and Biology Society, Amsterdam, 1996, pp. 144-145. cited by applicant

Maguire et al., "Evaluation of Intrapedicular Screw Position Using Intraoperative Evoked Electromyography," *Spine*, 1995, 20(9): 1068-1074. cited by applicant

Martin et al. "Initiation of Erection and Semen Release by Rectal Probe Electrostimulation (RPE)," *The Journal of Urology*, The Williams& Wilkins Co., 1983, 129: 637-642. cited by applicant

Minahan et al., "The Effect of Neuromuscular Blockade on Pedicle Screw Stimulation Thresholds" *Spine*, Department of Neurology, Johns Hopkins University School of Medicine, Oct. 1, 2000, 25(19): 2526-2530. cited by applicant

Pither et al., "The Use of Peripheral Nerve Stimulators for Regional Anesthesia: Review of Experimental Characteristics Technique and Clinical Applications," *Regional Anesthesia*, 1985, 10:49-58. cited by applicant

Raj et al., "Infraclavicular Brachial Plexus Block—A New Approach" *Anesthesia and Analgesia*, 1973, (52)6: 897-904. cited by applicant

Raj et al., "The Use of Peripheral Nerve Stimulators for Regional Anesthesia," *Clinical Issues in Regional Anesthesia*, 1985, 1(4):1-6. cited by applicant

Raj et al., "Use of the Nerve Stimulator for Peripheral Blocks," *Regional Anesthesia*, Apr.-Jun. 1980, pp. 14-21. cited by applicant

Raymond et al., "The Nerve Seeker: A System for Automated Nerve Localization," *Regional Anesthesia*, 1992, 17(3): 151-162. cited by applicant

Shafik, "Cavernous Nerve Simulation through an Extrapelvic Subpubic Approach: Role in Penile Erection," *Eur. Urol*, 1994, 26: 98-102. cited by applicant

Toleikis et al., "The Usefulness of Electrical Stimulation for Assessing Pedicle Screw Replacements," *Journal of Spinal Disorder*, 2000, 13(4): 283-289. cited by applicant

Medtronic Sofamor Danek "Union™/ Union-L™ Anterior & Lateral Impacted Fusion Devices:

Surgical Technique” Medtronic Sofamor Danek, 2001, 20 pages. cited by applicant

Bergey et al., “Endoscopic Lateral Transpsoas Approach to the Lumbar Spine,” *Spine*, 2004, 29(15): 1681-1688. cited by applicant

Dezawa et al., “Retroperitoneal Laparoscopic Lateral Approach to the Lumbar Spine: A New Approach, Technique, and Clinical Trial,” *Journal of Spinal Disorders*, 2000, 13(2): 138-143. cited by applicant

Gardocki, “Tubular discectomy minimizes collateral damage: A logical progression moves spine surgery forward,” *AAOS Now*, 2009, 5 pages. cited by applicant

Hovorka et al., “Five years' experience of retroperitoneal lumbar and thoracolumbar surgery,” *Eur Spine J.*, 2000, 9(1): S30-S34. cited by applicant

Kossmann et al., “The use of a retractor system (SynFrame) for open, minimal invasive reconstruction of the anterior column of the thoracic and lumbar spine,” *Eur Spine J.*, 2001, 10: 396-402. cited by applicant

Mayer, “A New Microsurgical Technique for Minimally Invasive Anterior Lumbar Interbody Fusion,” *Spine*, 1997, 22(6): 691-699. cited by applicant

Mayer, “The ALIF Concept,” *Eur Spine J.*, 2000, 9(1): S35-S43. cited by applicant

Mayer and Wiechert, “Microsurgical Anterior Approaches to the Lumbar Spine for Interbody Fusion and Total Disc Replacement,” *Neurosurgery*, 2002, 51(2): 159-165. cited by applicant

McAfee et al., “Minimally Invasive Anterior Retroperitoneal Approach to the Lumbar Spine: Emphasis on the Lateral BAK,” *Spine*, 1998, 23(13): 1476-1484. cited by applicant

Rao, et al. “Dynamic retraction of the psoas muscle to expose the lumbar spine using the retroperitoneal approach,” *J. Neurosurg Spine*, 2006, 5: 468-470. cited by applicant

Wolfla et al., “Retroperitoneal lateral lumbar interbody fusion with titanium threaded fusion cages,” *J. Neurosurg (Spine 1)*, 2002, 96: 50-55. cited by applicant

Larson and Maiman, “Surgery of the Lumbar Spine,” Thieme Medical Publishers, Inc., 1999, pp. 305-319. cited by applicant

Medtronic XOMED Surgical Products, Inc., NIM-Response Nerve Integrity Monitor Intraoperative EMG Monitor User's Guide, Revision B, 2000, 47 pages. cited by applicant

“Nu Vasive's spine surgery system cleared in the US,” *Pharm & Medical Industry Week*, Dec. 10, 2001, 1 page. cited by applicant

Pimenta, “Initial Clinical Results of Direct Lateral, Minimally Invasive Access to the Lumbar Spine for Disc Nucleus Replacement Using a Novel Neurophysiological Monitoring System.” *The 9.SUP.th./MAST*, May 2002, 1 page. cited by applicant

Pimenta et al., “The Lateral Endoscopic Transpsoas Retroperitoneal Approach (Letra) for Implants in the Lumbar Spine,” *World Spine II—Second Interdisciplinary Congress on Spine Care*, Aug. 2003, 2 pages. cited by applicant

Crock, H.V. MD., “Anterior Lumbar Interbody Fusion,” *Clinical Orthopaedics and Related Research*, No. One Hundred Sixty Five, 1982, pp. 157-163, 13 pages. cited by applicant

Mayer and Brock, “Percutaneous endoscopic discectomy: surgical technique and preliminary results compared to microsurgical discectomy,” *J. Neurosurg*, 1993, 78: 216-225. cited by applicant

Schaffer and Kambin, “Percutaneous Posterolateral Lumbar Discectomy and Decompression with a 6.9-Millimeter Cannula,” *The Journal of Bone and Joint Surgery*, 1991, 73A(6): 822-831. cited by applicant

Friedman, “Percutaneous discectomy: An alternative to chemonucleolysis,” *Neurosurgery*, 1983, 13(5): 542-547. cited by applicant

Brau, “Chapter 22: Anterior Retroperitoneal Muscle-Sparing approach to L2-S1 of the Lumbar Spine,” *Surgical Approaches to the Spine*. Robert G. Watkins, MD. (ed) 2003. pp. 165-181. cited by applicant

Kossmann et al., “Minimally Invasive Vertebral Replacement with Cages in Thoracic and Lumbar Spine,” *European Journal of Trauma*, 2001, 27: 292-300. cited by applicant

Mayer H. M. (ed.) Minimally Invasive S12ine Surgery: A Surgical Manual. 2000. 51 pages. cited by applicant

Pimenta et al., “Implante de protese de nucleo pulpost: analise inicial,” *Journal Brasileiro de Neurocirurgia*, 2001, 12(2): 93-96. cited by applicant

Traynelis, “Spinal Arthroplasty,” *Neurological Focus*, 2002, 13(2): 12 pages. cited by applicant

Zdeblick, Thomas A. (ed.). *Anterior Approaches to the Spine*. 1999. 43 pages. cited by applicant

CoRoent™ Marketing Brochure (9004001 A.0), NuVasive, Inc., 2004, 2 pages. cited by applicant

CoRoent™ Marketing Brochure (9004001 C.0), NuVasive, Inc., 2005, 2 pages. cited by applicant

CoRoent™ XL & XLR Marketing Brochure (9004225 AO), NuVasive, Inc., 2005, 2 pages. cited by applicant

CoRoent® XL & XLR Marketing Brochure (9004225 B.0), NuVasive, Inc., 2006, 2 pages. cited by applicant

CoRoent® XL & XLR Marketing Brochure (9004225 C.0), NuVasive, Inc., 2007, 2 pages. cited by applicant

CoRoent® XL Marketing Brochure (9500039 AO), NuVasive, Inc., 2006, 8 pages. cited by applicant

Baulot et al., “Complementary anterior spondylodesis by thoracoscopy. Technical note regarding an observation”, *Lyon Sur2.*, 90(5):347-351 (1994). cited by applicant

Berry et al., “A morphometric study of human lumbar and selected thoracic vertebrae, study of selected vertebrae” *S12ine* 12(4):362-367 (1996). cited by applicant

Crock, H. V., “A Short Practice of Spinal Surgery”, Second, revised edition, published by Springer-Verlag/Wein, New York (1993). cited by applicant

Edeland, H.G., “Some additional suggestions for an intervertebral disc prosthesis”, *Journal of Biomedical Engineering*, 7:57-62 (1985). cited by applicant

Kemp, H.B. S., “Anterior fusion of the spine for infective lesions in adults”, *Journal of Bone & Joint Surgery*, 55B(4):715-734 (1973). cited by applicant

Alleyne et al., “Current and future approaches to lumbar disc surgery: A literature review”, *Medsca12e Ortho12edics & S12orts Medicine*, 1, [www.medscape.com/Medscape/OrthoSportsMed/1997/vO1.n1 1/. . /mos3057], (1997). cited by applicant

Benini et al., “Undercutting decompression and posterior fusion with translaminar facet screw fixation in degenerative lumbar spinal stenosis: Technique and results”, *Neuro-Ortho12edics*, 17/18, 159-172 (1995). cited by applicant

Kambin, et al., “History and current status of percutaneous arthroscopic disc surgery”, *S12ine*, 21(24S):57S-61S (1996). cited by applicant

Stein et al., “Percutaneous facet joint fusion: Preliminary experience”, *Journal of Vascular and Interventional Radiology*, 4:69-74 (1993). cited by applicant

Vamvanu et al., “Surgical treatment of internal disc disruption: An outcome study of four fusion techniques”, *Journal of S12inal Disorders*, 11(5):375-382 (1998). cited by applicant

Damasceno et al., *Lumbar Lordosis: A Study of Angle Values and of Vertebral Bodies and Intervertebral Discs Role*, *Acta Ortol Bras*, 12(4):193-198 (2006). cited by applicant

Primary Examiner: Robert; Eduardo C

Assistant Examiner: Green; Michelle C

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending U.S. patent application Ser. No. 16/545,821 filed Aug. 20, 2019, which is a continuation of U.S. patent application Ser. No. 15/702,685 filed Sep. 12, 2017 (now U.S. Pat. No. 10,426,627), which is a continuation of U.S. patent application Ser. No. 14/918,137 filed Oct. 20, 2015 (now U.S. Pat. No. 9,757,246), which is a continuation of U.S. patent application Ser. No. 14/578,215 filed Dec. 19, 2014 (now U.S. Pat. No. 9,192,482), which is a continuation of U.S. patent application Ser. No. 13/653,335 filed Oct. 16, 2012 (now U.S. Pat. No. 8,920,500), which is a continuation of U.S. patent application Ser. No. 12/799,021 filed Apr. 16, 2010 (now U.S. Pat. No. 8,287,597), which claims the benefit of priority from U.S. Provisional Patent Application Ser. No. 61/212,921 filed Apr. 16, 2009, and U.S. Provisional Patent Application Ser. No. 61/319,823 filed Mar. 31, 2010, the entire contents of which is hereby expressly incorporated by reference into this disclosure as if set forth in its entirety herein.

FIELD

(1) The present invention relates to implants and methods for adjusting sagittal imbalance of a spine.

BACKGROUND

(2) A human spine has three main regions—the cervical, thoracic, and lumbar regions. In a normal spine, the cervical and lumbar regions have a lordotic (backward) curvature, while the thoracic region has a kyphotic (forward) curvature. Such a disposition of the curvatures gives a normal spine an S-shape. Sagittal imbalance is a condition in which the normal alignment of the spine is disrupted in the sagittal plane causing a deformation of the spinal curvature. One example of such a deformity is “flat-back” syndrome, wherein the lumbar region of the spine is generally linear rather than curved. A more extreme example has the lumbar region of the spine exhibiting a kyphotic curvature such that the spine has an overall C-shape, rather than an S-shape. Sagittal imbalance can be a problem in that it is biomechanically disadvantageous and generally results in discomfort, pain, and an awkward appearance in that the patient tends to be bent forward excessively.

(3) Various treatments for sagittal imbalance are known in the art. These treatments generally involve removing at least some bone from a vertebra (osteotomy) and sometimes removal of the entire vertebra (vertebrectomy), in order to reduce the posterior height of the spine in the affected region and recreate lordotic curve. Such procedures are traditionally performed via an open, posterior approach involving a large incision (often to expose multiple spinal levels at the same time) and require stripping of the muscle tissue away from the bone. These procedures can have the disadvantages of a large amount of blood loss, high risk, and a long and painful recovery for the patient. Furthermore, depending upon the patient, multiple procedures, involving both anterior and posterior approaches to the spine, may be required.

(4) The present invention is directed at overcoming, or at least improving upon, the disadvantages of the prior art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many advantages of the present invention will be apparent to those skilled in the art with a reading of this specification in conjunction with the attached drawings, wherein like reference numerals are applied to like elements and wherein:

(2) FIG. 1 is a lateral view representing a portion of a sagittally imbalanced lumbar spine lacking the normal lordotic curvature;

(3) FIG. 2 is a side view representing the lumbar spine of FIG. 1 after restoration of the lordotic curvature using a hyper-lordotic fusion implant, according to one example embodiment;

- (4) FIG. 3 is a top down view depicting the creation of a lateral access corridor formed with surgical access system through the side of the patient to the target disc space, according to one example embodiment;
- (5) FIG. 4 is a perspective view depicting a lateral access corridor formed with a retractor assembly through the side of the patient to the target disc space, according to one example embodiment;
- (6) FIG. 5 is a front perspective view of an Anterior Longitudinal Ligament (ALL) resector for safely releasing the ALL through a lateral access corridor, according to one example embodiment;
- (7) FIG. 6 is an enlarged front perspective view of the distal portion of the ALL resector of FIG. 5;
- (8) FIG. 7 is an enlarged side view of the distal portion of the ALL resector of FIG. 5;
- (9) FIG. 8 is an exploded front perspective view of the ALL resector of FIG. 5;
- (10) FIG. 9 is a front perspective view of an ALL resector for safely releasing the ALL through a lateral access corridor, according to another example embodiment;
- (11) FIG. 10 is a side view of the ALL resector of FIG. 9;
- (12) FIG. 11 is an enlarged side view of the distal end of the ALL resector of FIG. 9;
- (13) FIG. 12 is an enlarged side view of the distal end of the ALL resector of FIG. 9 having curved finger portions;
- (14) FIGS. 13-15 are anterior-front perspective, posterior-rear perspective, and front views of a hyper-lordotic lateral implant according to one example embodiment;
- (15) FIGS. 16-17 are posterior-rear perspective and rear views of a hyper-lordotic lateral implant according to another example embodiment; and
- (16) FIGS. 18-19 are posterior-rear perspective and rear views of a hyper-lordotic lateral implant according to still another example embodiment.

DESCRIPTION OF A PREFERRED EMBODIMENT

(17) Illustrative embodiments of the invention are described below. In the interest of clarity, not all features of an actual implementation are described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure. The methods and devices described herein include a variety of inventive features and components that warrant patent protection, both individually and in combination.

(18) With reference to FIGS. 1-2, devices and methods of the present invention are utilized to correct sagittal imbalance, including lumbar kyphosis, by increasing the anterior height of the affected spinal area (as opposed to reducing the posterior height). FIG. 1 illustrates a portion of the lumbar spine lacking the standard lordotic curvature. To correct the sagittal imbalance, illustrated in FIG. 2, a hyper-lordotic implant 10 is positioned into the disc space at the appropriate spinal level (e.g. between V1 and V2). An anterior sidewall 12 of hyper-lordotic implant 10 has a height significantly larger than an opposing posterior sidewall 14 such that when the implant is positioned within the disc space the anterior aspects of V1 and V2 are forced apart while the posterior aspects are not (or at least not to the same degree), thus imparting a lordotic curvature into the spine. To allow the anterior aspects of V1 and V2 to separate and receive the hyper-lordotic implant 10, the Anterior Longitudinal Ligament (ALL) that runs along the anterior aspect of the spine may be released or cut (15). According to a preferred method, the implant 10 is implanted through a lateral access corridor formed through the side of the patient. Accessing the targeted spinal site through the lateral access corridor avoids a number of disadvantages associated with posterior access (e.g. cutting through back musculature and possible need to reduce or cut away part of the posterior bony structures like lamina, facets, and spinous process) and anterior access (e.g. use of an access surgeon to move various organs and blood vessels out of the way in order to reach the target site).

Accordingly, by accessing the target site via a lateral access approach and correcting the sagittal imbalance without reducing the posterior height (i.e. no bone removal) the high blood loss and painful recovery associated previous methods may be avoided (or at least mitigated).

(19) According to one example, the lateral access approach to the targeted spinal space may be performed according to the instruments and methods described in commonly owned U.S. Pat. No. 7,207,949 entitled "Surgical Access System and Related Methods," and/or U.S. patent application Ser. No. 10/967,668 entitled "Surgical Access System and Related Methods," the entire contents of which are each incorporated herein by reference as if set forth herein in their entireties. With reference to FIGS. 3-4, a discussion of the lateral access instruments and methods is provided in brief detail. With the patient **20** positioned on their side, a surgical access system **26** is advanced through an incision **20**, into the retroperitoneal space **22**, and then through the psoas muscle **24** until the targeted spinal site (e.g. the disc space between V1 and V2) is reached. The access system **26** may include at least one tissue dilator, and preferably includes a sequential dilation system **28** with an initial dilator **30** and one or more additional dilators **32** of increasing diameter, and a tissue retractor assembly **34**. As will be appreciated, the initial dilator **30** is preferably advanced to the target site first, and then each of the additional dilators **32** of increasing diameter are advanced in turn over the previous dilator. A k-wire (not shown) may be advanced to the target site and docked in place (for example, by inserting the k-wire into the vertebral disc) prior to, in concurrence with, or after advancing the initial dilator **30** to the target site. With the sequential dilation system **28** positioned adjacent the target site (and optionally docked in place via the k-wire), the retractor assembly **34** is then advanced to the target site over the sequential dilation system **28**. According to the embodiment shown, the retractor assembly **34** includes retractor blades **36**, **38**, **40** and a body **42**. With the sequential dilation system **28** removed, the retractor blades **36**, **38**, and **40** are separated (FIG. 5), providing the lateral access corridor through which instruments may be advanced to prepare the disc space and insert the implant **10**. According to one example, the posterior blade **36** may be fixed in position relative to the spine prior to opening the retractor blades. This may be accomplished, for example by attaching a shim to the blade **36** (e.g. via track **44** including dove tail grooves formed on the interior of blade **36**) and inserting the distal end of the shim into the disc space. In this manner, the posterior blade **36** will not move posteriorly (towards nerve tissue located in the posterior portion of the psoas muscle). Instead, the blades **38** and **40** will move away from the posterior blade **36** to expand the access corridor. Additionally, nerve monitoring (including determining nerve proximity and optionally directionality) is preferably performed as each component of the access system **26** is advanced through the psoas muscle, protecting the delicate nerve tissue running through the psoas, as described in the '949 patent and '668 application. Monitoring the proximity of nerves also allows the posterior blade **36** of the retractor assembly **34** to be positioned very posterior (all the way back to the exiting nerve roots), thus exposing a greater portion of the disc space than would otherwise be safely achievable. This in turn permits full removal of the disc and implantation of a wider implant and the wider implant makes utilization of the hyper-lordotic implant with a large lordotic angle (e.g. between 20-40 degrees) more practical.

(20) With the lateral access corridor formed (as pictured in FIG. 5) the target site may be prepped for insertion of the implant **10**. Preparation of the disc space may include performing an annulotomy, removal of disc material, and abrasion of the endplates and instruments such as annulotomy knives, pituitaries, curettes, disc cutters, endplate scrapers may be used. Additionally, as discussed above, it may generally be necessary to release the ALL in order to create enough flexibility between the adjacent vertebrae (e.g. V1 and V2) to receive the hyper-lordotic implant **10**. Unlike an anterior approach (where the great vessels and other tissue lying anterior to the disc space is retracted during the approach), when the target disc is approached laterally, the great vessels remain adjacent to the ALL along the anterior face of the spine. Thus, while cutting the ALL is generally simple and necessary during an anterior approach surgery, cutting the ALL during

a lateral approach surgery has typically been unnecessary and can be difficult because of the need to avoid damaging the great vessels. Accordingly, FIGS. 5-14 set forth various example embodiments of ALL resecting instruments for safely releasing the ALL from a lateral approach.

(21) FIGS. 5-8 illustrate one example ALL resector **50**. The ALL resector **50** includes a tissue retractor **52** and a sliding blade **68** and functions to both sever the ALL and protect surrounding tissue (e.g. great vessels, nerves etc. . . .) from unwanted damage while the ALL is being severed. The tissue retractor **52** includes a handle **54**, hollow shaft **56**, and head **58**. The head **58** is curved, preferably such that the inside surface **60** complements the curvature of the anterior aspects of the spinal target site. The head **58** may thus be positioned through the lateral access corridor to the spine and such that the curved interior surface **60** nestles around the curved anterior aspect of the spine. The outside surface **62** will form a barrier, protecting tissue along the anterior spine from inadvertent contact with the sliding blade when the ALL is cut. Furthermore, the tissue retractor **52** can be further manipulated move tissue and further expose the anterior aspect of the target site. The hollow shaft **56** includes a central lumen **64** with an opening adjacent the head **58** and another opening at the opposing end such that the sliding blade **54** may travel through the shaft **56**.

(22) The sliding blade **68** includes a blade **70** that is secured to the distal end of an extender **72** by way of an attachment feature **74**. The attachment feature **74** as shown is similar to known attachment features used for attaching a blade at the end of a scalpel. It will be appreciated that any number of mechanisms may be used to attach blade **70** to extender **72**. Blade **70** may be disposable and extender **72** may be reusable. Alternatively, both blade **70** and extender **72** may be reusable or both may be disposable. The blade **70** includes a cutting edge **76** that, when advanced beyond the lumen **64** of shaft **56**, cuts through tissue or material situated adjacent the cutting edge **76**.

(23) The proximal end of the extender **72** includes a grip **78** that a user may use to manipulate the position of the sliding blade **40** relative to the shaft **56** and head **58**. At least one stop feature **80** extends from the outer surface of the extender **72** which engages with a track **66** that extends along a portion of the elongated shaft **56**. The track **66** limits the longitudinal travel of the sliding blade **68** relative to the shaft **56** so that the sliding blade **68** remains slidably mated to the tissue retractor **52** without becoming unassembled and such that the blade **70** cannot extend beyond the protective head **58**. Additionally, the stop feature **80** restricts rotation of the sliding blade **68** relative to the tissue retractor **52**.

(24) FIGS. 9-11 illustrate another example embodiment of an ALL resector **82**. The ALL resector **82** includes a handle **84** (for example, a T-handle) located at the proximal end of the elongated shaft **86** and a distal head **88** for resecting the ALL. The distal head **88** includes distally extending first and second fingers **90**, **92**, which form an opening **94** therebetween. First and second tapered surfaces **96**, **98** which extend a distance from the elongated shaft **84** along the fingers **90**, **92** enable the distal head **88** to insert gently between tissue. As best shown in FIG. 11, the first finger **90** may be shorter in length than the second finger **92**. However, the first and second finger **90**, **92** may be provided in any number of length configurations without departing from the scope of the present invention. By way of example, it has been contemplated that the first finger **14** may be completely removed. Alternatively, as pictured in FIG. 12, the fingers **90** and **92** may be curved and have a more substantial width than shown in FIGS. 9-11. The curvature of the first and second fingers allows the distal head **88** to follow closely along the anterior side of the spine and/or along a curved spatula (not shown) positioned adjacent the anterior side of the vertebral body. Though not shown, a user may optionally insert a spatula along the anterior portion of the ALL prior to inserting the ALL retractor **82**. The spatula may serve as additional protection between the delicate tissue anterior to the ALL and the cutting blade **97** of the ALL resector **82**. With a spatula in place the user may insert the distal head **88** such that it approaches the lateral side of the ALL and is guided along the inside edge of the spatula. By way of example, the spatula may be straight or curved to match the selected fingers of the distal head **88**.

(25) A cutting blade **97** is exposed between the first and second fingers **90**, **92** in the opening **17**. A

slot **99** formed along a side of the distal head **88** allows a cutting blade **97** to be inserted and removed from the distal head **88**. Thus, the cutting blade **97** may be disposable and the remainder of the ALL resector **82** may be reusable. Alternatively, both cutting blade **97** and remainder of the ALL resector **82** may be reusable or both may be disposable. In use, the ALL resector **82** is preferably positioned such that the second finger **92** is aligned along the anterior side of the ALL and the first finger **90** is aligned along the posterior side of the ALL, thus, at least partially bounding the ALL on either side. The ALL resector **82** is advanced forward so that the cutting blade **97** cuts through the ALL from one lateral edge to the other. As discussed above, the second finger **92** is preferably aligned along the anterior side of the ALL as the distal head **88** is advanced, thereby shielding the tissue lying anterior to the finger **92** (e.g. great vessels, etc.) from the cutting blade **97**. Furthermore, as the user advances the ALL resector **82**, the fingers **90**, **92** may also act as a stabilizing guide.

(26) Turning now to FIGS. **13-19**, various embodiments of the hyper-lordotic implant **10** are described. FIGS. **13-15**, for example, illustrate an example implant **110**. Implant **110** may preferably be comprised of any suitable non-bone composition having suitable radiolucent characteristics, including but not limited to polymer compositions (e.g. poly-ether-ether-ketone (PEEK) and/or poly-ether-ketone-ketone (PEKK)) or any combination of PEEK and PEKK. Other materials such as for example, metal, ceramics, and bone may also be utilized for the implant **110**. Implant **110** has a top surface **116** and bottom surface **118** for contacting V1 and V2, anterior sidewall **112**, posterior sidewall **114**, and front or leading side **120**, and rear or trailing side **122**. As discussed, the anterior sidewall **112** has a height greater than the posterior sidewall **114** such that the top surface **116** and bottom surface **118** converge towards each other in the posterior direction. By way of example, the top and bottom surfaces may converge at an angle between 20 and 40 degrees. It is contemplated that variations of the implant **110** may be simultaneously provided such that the user may select from different available ranges. For example, variations may be provided with 20 degree, 30 degree, and 40 degree angles. The top and bottom surfaces may be planar or provided as convex to better match the natural contours of the vertebral end plates. The length of the implant **110** is such that it may span from one lateral aspect of the disc space to the other, engaging the apophyseal ring on each side. By way of example, the implant **110** may be provided with a length between 40 mm and 60.

(27) Having been deposited in the disc space, the implant **110** facilitates spinal fusion over time by maintaining the restored curvature as natural bone growth occurs through and/or past the implant **10**, resulting in the formation of a boney bridge extending between the adjacent vertebral bodies.

(28) The implant **110** may be provided with any number of additional features for promoting fusion, such as apertures **127** extending between the top and bottom surfaces **116-117** which allow a boney bridge to form through the implant **110**. Various osteoinductive materials may be deposited within the apertures **127** and/or adjacent to the implant **110** to further facilitate fusion. Such osteoinductive materials may be introduced before, during, or after the insertion of the exemplary spinal fusion implant **110**, and may include (but are not necessarily limited to) autologous bone harvested from the patient receiving the spinal fusion implant, bone allograft, bone xenograft, any number of non-bone implants (e.g. ceramic, metallic, polymer), bone morphogenic protein, and bio-resorbable compositions, including but not limited to any of a variety of poly (D,L-lactide-co-glycolide) based polymers. Visualization apertures **128** situated along the sidewalls, may aid in visualization at the time of implantation and at subsequent clinical evaluations. More specifically, based on the generally radiolucent nature of the preferred embodiment of implant **110**, the visualization apertures **128** provide the ability to visualize the interior of the implant **110** during X-ray and/or other suitable imaging techniques. Further, the visualization apertures **128** will provide an avenue for cellular migration to the exterior of the implant **110**. Thus the implant **110** will serve as additional scaffolding for bone fusion on the exterior of the implant **110**.

(29) The implant **110** also preferably includes anti-migration features designed to increase the

friction between the implant **110** and the adjacent contacting surfaces of the vertebral bodies V1 and V2 so as to prohibit migration of the implant **110** after implantation. Such anti-migration features may include ridges provided along the top surface **116** and/or bottom surface **118**. Additional anti-migration features may also include spike elements **124** disposed along the top and bottom surfaces.

(30) Tapered surfaces **126** may be provide along the leading end **120** to help facilitate insertion of the implant **110**. Additional instrumentation may also be used to help deploy the implant **110** into the disc space. By way of example, the implant installation device shown and described in detail in the commonly owned and co-pending U.S. patent application Ser. No. 12/378,685, entitled "Implant Installation Assembly and Related Methods," filed on Feb. 17, 2009, the entire contents of which is incorporated by reference herein, may be used to help distract the disc space and deposit the implant therein.

(31) FIGS. **16-17** illustrate an implant **210** according to another example embodiment. The implant **210** share many similar features with the implant **110** such that like feature are numbered alike and repeat discussion in not necessary. The implant **210** differs from the implant **110** in that a trailing side **212** is configured for fixed engagement to one of the adjacent vertebral bodies (i.e. V1 or V2) to supplement the anti migration features and ensure the hyper-lordotic implant is not projected out of the disc space. Specifically, the implant **210** includes a tab **213** extending vertically above the top surface **116** and/or below the bottom surface **118**. Tabs **213** facilitate fixation of the implant **210** to the adjacent vertebrae. In the example shown in FIGS. **16-17**, the tab **213** extends past both the bottom and top surfaces **118** and **116**, respectively. It will be appreciated, however, that the implant **210** may include, without departing from the scope of the present invention and by way of example only, a tab **213** that extend solely in one direction. Additionally, multiple tabs **210** of various sizes and/or shapes may be used throughout a single implant **210** to facilitate implantation or fixation. In the illustrated example, the tab has a single fixation aperture **214** for enabling the engagement of a fixation anchor (not shown) within the vertebral bone to secure the placement of the implant. In use, when the implant **210** is positioned within the disc space, the tab **213** engages the exterior of the upper and lower vertebra and the fixation anchor may be driven into the side of either the upper or lower vertebra. One will appreciate that various locking mechanism may be utilized and positioned over or within the fixation aperture **214** to prevent the fixation anchor from unwanted disengagement with the implant **210**. For example, a suitable locking mechanism may be in the form of a canted coil disposed within the fixation aperture **214** or a screw that may be engaged to the trailing end **212** and cover all or a portion of the fixation aperture **214** after the fixation anchor positioned.

(32) FIGS. **18-19** illustrate an implant **310** according to another example embodiment. The implant **310** shares many similar features with the implants **110** and **110** such that like feature are numbered alike and repeat discussion in not necessary. The implant **310** differs from the implant **110** in that a trailing side **312** is configured for fixed engagement to one of the adjacent vertebral bodies (i.e., V1 or V2) to supplement the anti-migration features and ensure the hyper-lordotic implant is not projected out of the disc space. Specifically, the implant **310** includes a tab **313** extending vertically above the top surface **116** and/or below the bottom surface **118**. Tab **313** facilitates fixation of the implant **310** to the adjacent vertebrae. Tab **313** extends past both the bottom and top surfaces **118** and **116**, respectively. It will be appreciated, however, multiple tabs **310** of various sizes and/or shapes may be used throughout a single implant **310** to facilitate implantation or fixation. The implant **310** differs from the implant **210** in that the tab **213** has includes a pair of fixation apertures **214** for enabling the engagement of a pair fixation anchors (not shown) within the vertebral bone to secure the placement of the implant. In use, when the implant **310** is positioned within the disc space, the tab **313** engages the exterior of the upper and lower vertebra and a fixation anchor is driven into the side of each of the upper or lower vertebra. One will appreciate that various locking mechanism may be utilized and positioned over or within the fixation apertures **314** to prevent the

fixation anchors from unwanted disengagement with the implant **310**. One will also appreciate that single locking mechanism may be used to cover both fixation apertures **314**, or, a separate locking mechanism may be used to cover each fixation aperture **314**. Again, by way of example, a suitable locking mechanism may be in the form of a canted coil disposed within each fixation aperture **314** or a screw that may be engaged to the trailing end **312** and cover all or a portion of each fixation aperture **314** after the fixation anchors are positioned.

(33) While this invention has been described in terms of a best mode for achieving this invention's objectives, it will be appreciated by those skilled in the art that variations may be accomplished in view of these teachings without deviating from the spirit or scope of the invention. For example, particularly at L5-S1 where the pelvic bone makes a lateral access approach difficult, an antero-lateral approach similar to the approach utilized during appendectomies may be utilized.

Claims

1. A resector instrument comprising: a tissue retractor comprising: a handle, a hollow shaft coupled to the handle, the hollow shaft having a central lumen therein and defining a longitudinal axis, a first opening at a proximal end of the central lumen, a second opening at a distal end of the central lumen, a longitudinal track disposed on the hollow shaft, and a curved head adjacent the second opening; and a sliding blade disposed within the central lumen and configured to move axially therein, wherein the sliding blade includes a stop slidably coupled to the longitudinal track and configured to prevent rotation of the sliding blade relative to the hollow shaft, wherein the sliding blade is configured to sever tissue in an extended position, and the curved head is configured to protect surrounding tissue from contact with the sliding blade.
2. The resector instrument of claim 1, wherein the tissue is a ligament.
3. The resector instrument of claim 2, wherein the ligament is the Anterior Longitudinal Ligament (ALL).
4. The resector instrument of claim 2, wherein the extender further comprises a grip disposed at a proximal end thereof, the grip being configured to control an axial position of the sliding blade relative to the shaft and the curved head.
5. The resector instrument of claim 2, wherein rotation of the sliding blade is restricted relative to the tissue retractor.
6. The resector instrument of claim 5, further comprising: a track disposed on the hollow shaft and extending along a portion thereof; and a stop feature disposed on the extender and extending from an outer surface thereof, wherein the stop feature is configured to engage with the track, and wherein the track is configured to limit axial movement of the sliding blade relative to the hollow shaft.
7. The resector instrument of claim 1, wherein the sliding blade comprises: an extender; and a blade secured to a distal end of the extender, wherein the blade includes a distal cutting edge that, when advanced distally beyond the second opening, is configured to sever the tissue adjacent the cutting edge.
8. The resector instrument of claim 7, wherein the blade is configured to be disposable.
9. The resector instrument of claim 7, wherein the extender is configured to be reusable.
10. The resector instrument of claim 7, wherein the blade and the extender are configured to be reusable.
11. The resector instrument of claim 7, wherein the blade and the extender are configured to be disposable.
12. The resector instrument of claim 1, wherein an inner surface of the curved head is configured to complement a curvature of an aspect of a target site.
13. The resector instrument of claim 12, wherein the inner surface of the curved head is configured to complement a curved anterior aspect of a spine of a patient.

14. The resector instrument of claim 1, wherein an outer surface of the curved head is configured to form a barrier between the sliding blade and a non-target tissue, thereby protecting the non-target tissue from inadvertent contact with the sliding blade.
 15. The resector instrument of claim 14, wherein the non-target tissue is tissue along an anterior aspect of a spine of a patient.
 16. The resector instrument of claim 1, wherein the tissue retractor is configured to manipulate tissue and further expose an aspect of a target site.
 17. The resector instrument of claim 1, wherein the longitudinal track is configured to prevent the sliding blade from extending past a distal end of the curved head.
-